

Project Report: CodeQuest Study Report Tutor - Generative AI Engineer

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Family:	Generative AI Engineer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

RAG over study report JSON/Markdown; structured prompts return chapter summaries and quiz questions.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview *(Level: Intermediate)**

CodeQuest Study Report Tutor is a intermediate-level end-to-end project for Generative AI Engineer. Duration: 3-4 weeks. RAG over study report JSON/Markdown; structured prompts return chapter summaries and quiz questions. Tools: LangChain, FastAPI, OpenAI, Chroma.**

Code Reference:

```
Objective: Index Role_Reports catalog and chapters
Objective: Structured JSON output for quizzes
Objective: Rate limiting and cost tracking
Objective: Evaluation set with 20 benchmark questions
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Generative AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Generative AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Index [Intermediate]

Chapter focus: Phase: Index *(Level: Intermediate)**

Load report corpus. Complete these tasks: JSON + MD parsers; Metadata filters. Deliverable for this phase: Vector index.

Code Reference:

- [] JSON + MD parsers
- [] Metadata filters

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Index is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Vector index to the teacher.

Chapter 3: Phase: Prompts [Intermediate]

Chapter focus: Phase: Prompts *(Level: Intermediate)**

Tutor persona prompts. Complete these tasks: Summary template; Quiz template. Deliverable for this phase: Prompt library.

Code Reference:

- [] Summary template
- [] Quiz template

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Prompts is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Prompt library to the teacher.

Chapter 4: Phase: Eval [Intermediate]**Chapter focus: Phase: Eval** *(Level: Intermediate)**

20 benchmark Q&A. Complete these tasks: Accuracy scoring; Hallucination check. Deliverable for this phase: Eval spreadsheet.

Code Reference:

- [] Accuracy scoring
- [] Hallucination check

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Eval is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Eval spreadsheet to the teacher.

Chapter 5: Phase: API [Intermediate]**Chapter focus: Phase: API** *(Level: Intermediate)**

FastAPI wrapper. Complete these tasks: /summarize; /quiz. Deliverable for this phase: Live tutor API.

Code Reference:

- [] /summarize
- [] /quiz

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase API is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Live tutor API to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review** *(Level: Intermediate)

Submit all final deliverables: Tutor API; Eval report; Prompt library doc; Demo video. Skills demonstrated: RAG, Prompt engineering, Evaluation. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Tutor API
Deliverable: Eval report
Deliverable: Prompt library doc
Deliverable: Demo video
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Tutor API and passes the rubric.